FINAL YEAR PROJECT

BSc Mathematical Sciences

Modelling fluid flow to measure wind turbine energy efficiency

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Introduction

The goal of this Final Year Project has been to investigate the use of complex numbers to mathematically model the effects of the wind on the rotation of a wind turbine in idealised conditions. The surrounding airflow is considered for simplicity to be uniform, inviscid, time-independent and non-turbulent.

The project was first considered two-dimensionally, in the context of lift generation in an airfoil on the complex plane, before extending results to the three-dimensional case. Introductory airfoil was studied as in D. J. Acheson's *Elementary Fluid Dynamics*, L.M. Milne-Thomson's *Theoretical Aerodynamics* and John. D Anderson's *Fundamentals of Aerodynamics*.

Airfoil Theory

Flow Around a Circle

Before examining flows around airfoils or other objects, we look at the unobstructed flow. For our purposes this is a steady uniform airflow at some velocity U at an angle α to the x-axis, defined as

$$\vec{u}(x, y) = (U\cos\alpha, U\sin\alpha, 0).$$

The velocity potential ϕ and streamfunction ψ are easily calculable for this flow. Defining $z = x + iy$, where $i^2 = -1$ as usual, allows a new quantity $w(z)$ to be defined, known as the **complex potential**:

$$w(z) = \phi + i\psi.$$

This function is used in the **Milne-Thomson Circle Theorem**:

For a fluid flow with complex potential $f(z)$ with singularities only at $|z - z_0| > a$,

$$w(z) = f(z - z_0) + f\left(\frac{a^2}{(z - z_0)^*}\right)^*.$$

where f^* and z^* are the complex conjugates of f and z respectively.

This corresponds to placing a circle of radius a into the flow, centered at z_0. The theorem allows us to solve for the complex potential of a flow around a circle if the unobstructed flow behaviour is known. After superimposing an irrotational vortex on the flow with circulation strength Γ (which will be necessary in the context of a tapered airfoil), sharing the circle's center, the complex potential is

$$w(z) = U\left((z - z_0)e^{-i\alpha} + \frac{a^2}{z - z_0}e^{i\alpha}\right) + \frac{i\Gamma}{2\pi}\ln(z - z_0).$$

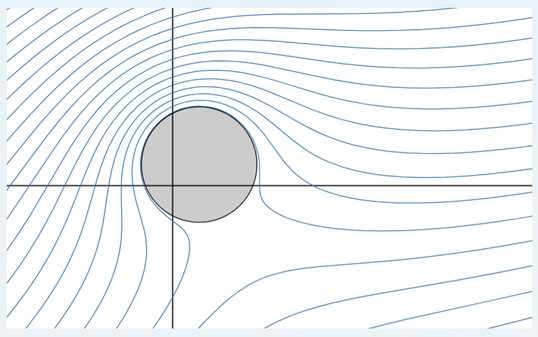

Figure 1: Streamlines for angled, left-to-right flow around a cylindrical profile with clockwise circulation.

Conformal Mapping - The Joukowsky Transform

To turn the equations used for a circular airfoil into something more suited for an airplane or wind turbine, a **conformal mapping** can be used. This is a function that's applied to the complex plane that *preserves angles between points.*

A well-known mapping for this purpose is the **Joukowsky Transform**, defined as

$$Z = z + \frac{b^2}{z},$$

where the b^2 term is an additional parameter that can be adjusted to create airfoil profiles with differing characteristics. Including the initial circle's radius a and center z_0, there are many degrees of freedom to play with when exploring transform outcomes. Under this transform, circles in the standard complex plane that intersect b on the real axis map to tapered airfoil profiles in the Z-plane, with sharp cusps at their trailing edges.

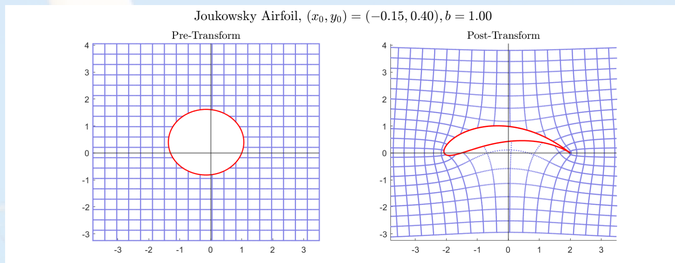

Figure 2: MATLAB plot visualising a circle and the surrounding space undergoing the Joukowsky transform with $b = 1$.

Applying a branch cut to the Z-plane along the real axis at $Z \in [-2b, 2b]$, the transform's inverse can be found to be

$$z = \frac{Z}{2} + \sqrt{\frac{1}{4}Z^2 - b^2}.$$

The complex potential for the flow around a circle can also undergo the Joukowsky transform, where every instance of z is substituted for the Joukowsky inverse. The resulting flow and its streamfunction can then be recovered.

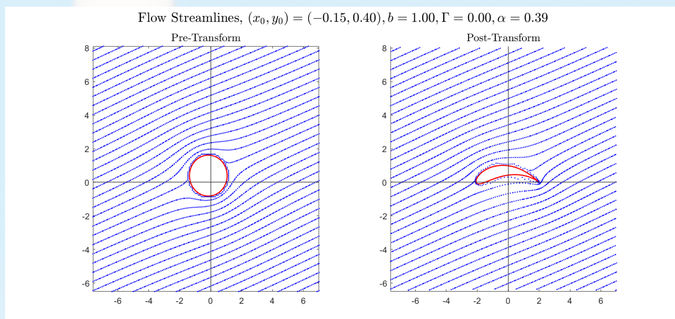

Figure 3: Diagram depicting streamlines for an angled uniform flow field around a cylindrical airfoil, which then undergoes the transform to simulate flow around a Joukowsky airfoil. Note that this flow has no superimposed vortex - this results in an unphysical stagnation point as fluid passes the airfoil.

Making use of the **Kutta Condition**, which states that fluid flowing past a thin airfoil should exit parallel to the airfoil's trailing edge, the added vortex's circulation strength Γ can be tuned to fit a given airfoil.

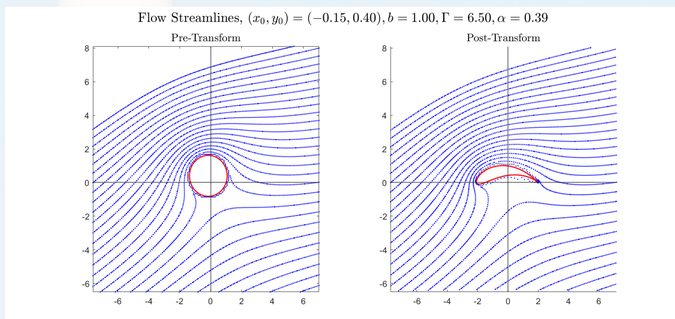

Figure 4: Diagram featuring the same flow field, whose circulation strength has been adjusted to satisfy the Kutta condition.

This downward deflection of fluid contributes to lift generation in wings, or torque generation in the turbine blade's case.

Calculating Forces

To find the forces acting on the airfoil section, we can apply **Blasius' Theorem**:

For a steady fluid flow in the complex plane with complex potential W and a body whose perimeter can be defined by a simple closed curve C, the sizes of the vertical and horizontal forces acting on the body - denoted F_x and F_y - satisfy

$$F_x - iF_y = \frac{1}{2}i\rho\oint_{\partial C}\left(\frac{dW}{dZ}\right)^2 dZ,$$

where ρ is the density of the fluid.

A special case of this theorem for thin airfoils, the **Kutta-Joukowsky Lift Theorem** gives the lift as

$$F_y = -\rho U\Gamma,$$

the same value of Γ that satisfies the Kutta condition.

Application to Turbines

These two-dimensional airfoil profiles can be layered together continuously to recreate the structure of a wind turbine blade. Values for the total torque T (force * length from fulcrum) acting on a blade of length L can be found by integrating across the length of the blade. Given that a typical blade widens and narrows as we move toward to the tip, this calculation could look like

$$T = -\rho\int_0^L x_l U(x_l)\Gamma(x_l)dx_l.$$

Γ varies with the turbines changing cross-section as the Kutta condition is to be satisfied for various cross-sectional shapes and sizes, and the fluid velocity U will be higher towards the tips, as well as increasing the faster the turbine rotates.

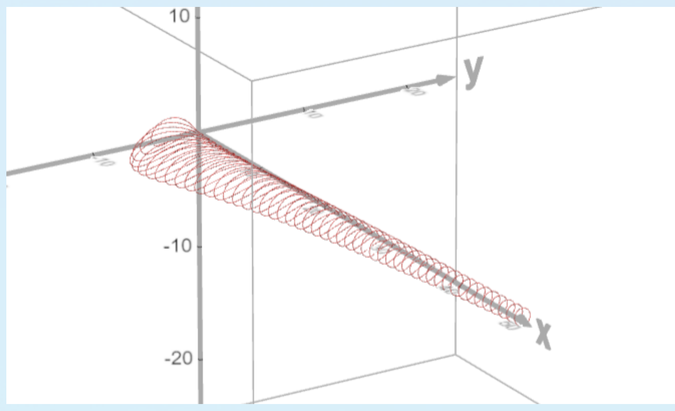

Figure 5: 3D plot of an example blade using variations of Joukowsky airfoils. the resulting slices have been shifted to line their trailing edges up.

Results and Conclusions

The assumptions needed to use the Kutta-Joukowsky Lift Theorem fall apart if airfoil characteristics are pushed to extremes (i.e. allowing an airfoil's trailing edge to curve massively thus increasing the circulation strength needed in the Kutta condition), which means this type of work is probably not suited to optimising turbine blade shapes. However, the methodology used could aid in analysing existing turbines.

Additional Work

It's remained to be seen if these types of models are consistent with real-life measurements. A statistical analysis of turbine rotation speeds could be made, and compared to airfoil theory calculations. Considerations can also be made for cases in which boundary layer separation occurs, resulting in turbulent wakes past an airfoil's trailing edge.

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